```
#include<iostream>
using namespace std;

struct Car
{
    string model;
    int year;
    float price;

};
int main()
{

    Car quantity[3];

    for (int i = 0; i < 3; i++)
    {
        cout<<"Enter the model of car "<<i+1<<"  :";
        cin>>quantity[i].model;
        cout<<"Enter the year of car "<<i+1<<"  :";
        cin>>quantity[i].year;
        cout<<"Enter the price of car "<<i+1<<"  :";
        cin>>quantity[i].price;

        cout<<endl;
    }
```

```
    cout<<"Car with price above 50000 : "<<endl;

    for (int i = 0; i < 3; i++)
    {
        if(quantity[i].price>=500000)
        {

            cout<<"Model : "<<quantity[i].model<<endl;;
            cout<<"Year : "<<quantity[i].year<<endl;
            cout<<"Price : "<<quantity[i].model<<endl;

        }
    }

    return 0;




}
```

```
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS/Assignment
labtask4_Q1 && "/home/syeda-areej-e-fatima/Sem 2/OOPS
Enter the model of car 1  :corolla
Enter the year of car 1  :2019
Enter the price of car 1  :45000

Enter the model of car 2  :civic
Enter the year of car 2  :2013
Enter the price of car 2  :55000

Enter the model of car 3  :alto
Enter the year of car 3  :2000
Enter the price of car 3  :42000

Car with price above 50000 :
Model : civic
Year : 2013
Price : 55000
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS/Assignment
```

```cpp
#include<iostream>
using namespace std;

struct address
{
    string city;
    string country;
};

struct contact_info
{
    string name;
    long int phone_no;
    address ad;
};

int main()
{
    contact_info data[10];
    for (int i = 0; i < 10; i++)
    {
        cout<<"Enter the name of person "<<i+1<<" :";
        cin>>data[i].name;
        cout<<"Enter the phone no. of person "<<i+1<<" :";
        cin>>data[i].phone_no;
        cout<<"Enter the city of person "<<i+1<<" :";
        cin>>data[i].ad.city;
        cout<<"Enter the country of person "<<i+1<<" :";
        cin>>data[i].ad.country;

        cout<<endl;
    }

    cout<<"----Displaying data of 10 people-----"<<endl;

    for (int i = 0; i < 10; i++)
    {
        cout<<"Name of person "<<i+1<<" :"<<data[i].name<<endl;
        cout<<"Phone no. of person "<<i+1<<" :"<<data[i].phone_no<<endl;
        cout<<"City of person "<<i+1<<" :"<<data[i].ad.city<<endl;
        cout<<"Country of person "<<i+1<<" :"<<data[i].ad.country<<endl;

        cout<<endl;
    }



    return 0;
}
```

```
cd "/home/syeda-areej-e-fatima/Sem 2/OOPS/Assignment/" &
ssignment/"labtask4_Q2
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS$ cd "/home/sy
 && "/home/syeda-areej-e-fatima/Sem 2/OOPS/Assignment/"
Enter the name of person 1 :waleeja
Enter the phone no. of person 1 :23456784564
Enter the city of person 1 :peshawar
Enter the country of person 1 :pakistan

Enter the name of person 2 :shayan
Enter the phone no. of person 2 :34567894566
Enter the city of person 2 :mumbai
Enter the country of person 2 :india

----Displaying data of 10 people-----
Name of person 1 :waleeja
Phone no. of person 1 :23456784564
City of person 1 :peshawar
Country of person 1 :pakistan

Name of person 2 :shayan
Phone no. of person 2 :34567894566
City of person 2 :mumbai
Country of person 2 :india
```

```cpp
#include<iostream>
using namespace std;

struct camera
{
    int megapixels;

};

struct mobiles
{
    camera cam;
    long int  price;
};

mobiles cam_data[5];


void five_in_one(int size)
{
    long int price=0;
    int highest = 0;
    float sum = 0.0;
    float average= 1.0;
    int count = 0;

    cout<<"Displaying the information of all the mobiles ";
    for (int i = 0; i < size; i++)
    {
        cout<<"Megapixel for mobile "<<i+1<<" : "<<cam_data[i].cam.megapixels<<endl;

        cout<<"Price for mobile "<<i+1<<" : "<<cam_data[i].price<<endl;
    }
    cout<<endl;
```

```
void five_in_one(int size)

    cout<<"Displaying the information of all the mobiles ";
    for (int i = 0; i < size; i++)
    {
        cout<<"Megapixel for mobile "<<i+1<<" : "<<cam_data[i].cam.megapixels<<endl;

        cout<<"Price for mobile "<<i+1<<" : "<<cam_data[i].price<<endl;
    }
    cout<<endl;

    for (int i = 0; i < size; i++)
    {
        if(price < cam_data[i].price)
        {
            price = cam_data[i].price;
        }
    }
    cout<<"Most expensive price : "<<price<<endl;
    cout<<endl;


    for (int i = 0; i < size; i++)
    {
        if(highest < cam_data[i].cam.megapixels)
        {
            highest = cam_data[i].cam.megapixels;
        }
    }
    cout<<"Most highest camera megapixel : "<<highest<<endl;
    cout<<endl;
```

```
19  void five_in_one(int size)
58
59      for (int i = 0; i < size; i++)
60      {
61          sum += cam_data[i].price;
62      }
63      average = (float) sum/size;
64      cout<<"Average price : "<<average<<endl;
65      cout<<endl;
66
67
68      for (int i = 0; i < size; i++)
69      {
70          if(cam_data[i].cam.megapixels >= 50)
71          {
72              count++;
73          }
74      }
75
76      cout<<"Mobiles with camera >50 MP : "<<count<<endl;
77      cout<<endl;
78
79
80
81  }
82
83
84  int main()
85  {
86
87      for (int i = 0; i < 5; i++)
88      {
89
```

```
int main()
{

    for (int i = 0; i < 5; i++)
    {

        cout<<"Enter the megapixel for mobile (>0)"<<i+1<<" : ";
        cin>>cam_data[i].cam.megapixels;
        if(cam_data[i].cam.megapixels <= 0)
        {
            cout<<"You entered invalid megapixel "<<endl;
            cout<<"enter the megapixel for mobile (>0)"<<i+1<<" again : ";
            cin>>cam_data[i].cam.megapixels;

        }

        cout<<"Enter the price for mobile (>=0)"<<i+1<<" : ";
        cin>>cam_data[i].price;
        if(cam_data[i].cam.megapixels < 0)
        {
            cout<<"You entered invalid price "<<endl;
            cout<<"enter the price for mobile (>=0)"<<i+1<<" again : ";
            cin>>cam_data[i].price;
        }


    }

    five_in_one(5);


}
```

```
syeda-areej-e-fatima@velanora:~/Sem 2/OOPS$ cd "/home/syeda-areej-e-fatima/Sem
 && "/home/syeda-areej-e-fatima/Sem 2/OOPS/Assignment/"labtask4_Q3
Enter the megapixel for mobile (>0)1 : 23
Enter the price for mobile (>=0)1 : 45678
Enter the megapixel for mobile (>0)2 : 56
Enter the price for mobile (>=0)2 : 34534
Enter the megapixel for mobile (>0)3 : 45
Enter the price for mobile (>=0)3 : 56784
Enter the megapixel for mobile (>0)4 : 67
Enter the price for mobile (>=0)4 : 456780
Enter the megapixel for mobile (>0)5 : 87
Enter the price for mobile (>=0)5 : 345678
Displaying the information of all the mobiles Megapixel for mobile 1 : 23
Price for mobile 1 : 45678
Megapixel for mobile 2 : 56
Price for mobile 2 : 34534
Megapixel for mobile 3 : 45
Price for mobile 3 : 56784
Megapixel for mobile 4 : 67
Price for mobile 4 : 456780
Megapixel for mobile 5 : 87
Price for mobile 5 : 345678

Most expensive price : 456780

Most highest camera megapixel : 87

Average price : 187891

Mobiles with camera >50 MP : 3
```